B. Additional experiments for ICML 2026 rebuttals

B.1. Training, validation, and test performance during training

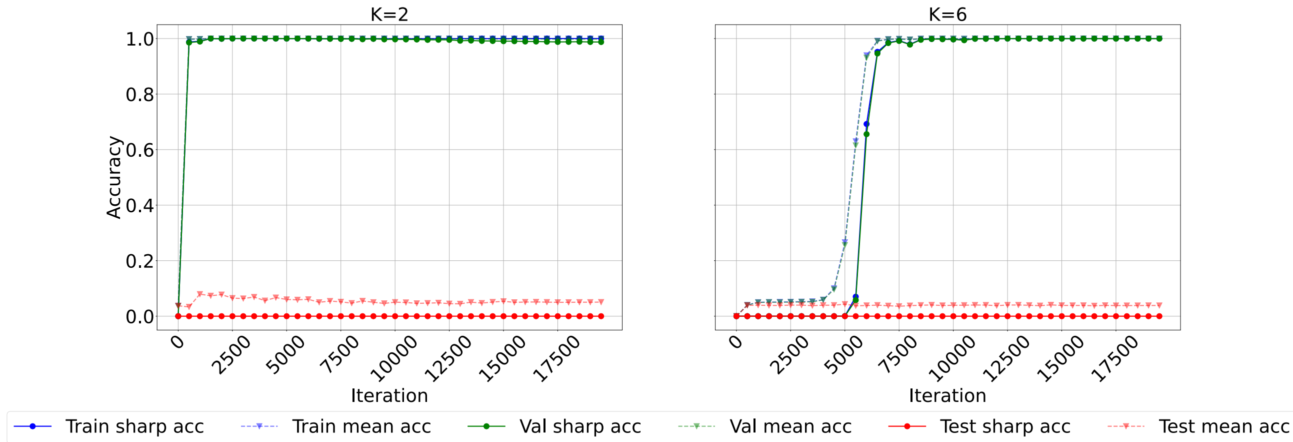

Figure 25. **Uniform benchmark (without identity), N_heads = 6, N_layers = 3, Direct (Relative)**: We observe that the models learn to predict the outputs of training compositions for both K=2 and for K=6, while the test accuracy on unseen compositions remains 0% throughout. We report two metrics: (1) **sharp accuracy**, where a prediction scores 1.0 only if the entire output sequence is correct and 0.0 otherwise; and (2) **mean token accuracy**, average accuracy of predictions in the output sequence, with both metrics averaged across compositions from each split. For reference, 100 epochs correspond to roughly 20k iterations (batch size 512, 100k training samples, 195 iterations per epoch).

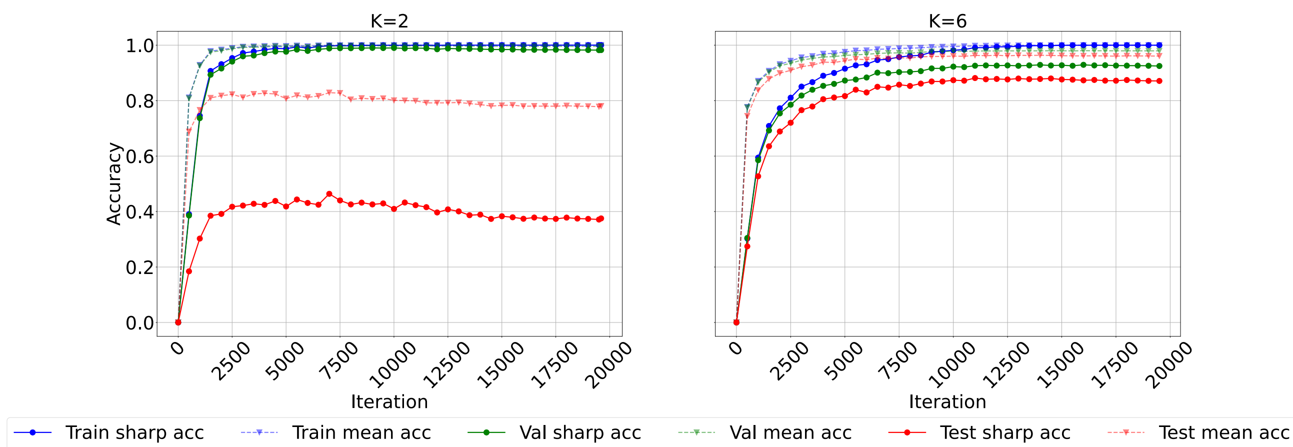

Figure 26. **Diverse benchmark, N_heads = 6, N_layers = 3, Direct (Relative)**: We observe non-trivial compositional generalization, as the models perform well on training and validation compositions, the test performance also increases. We observe better generalization for $K = 6$ than for $K = 2$ due to more training diversity and shared equivalences.

B.2. Sample efficiency ablations

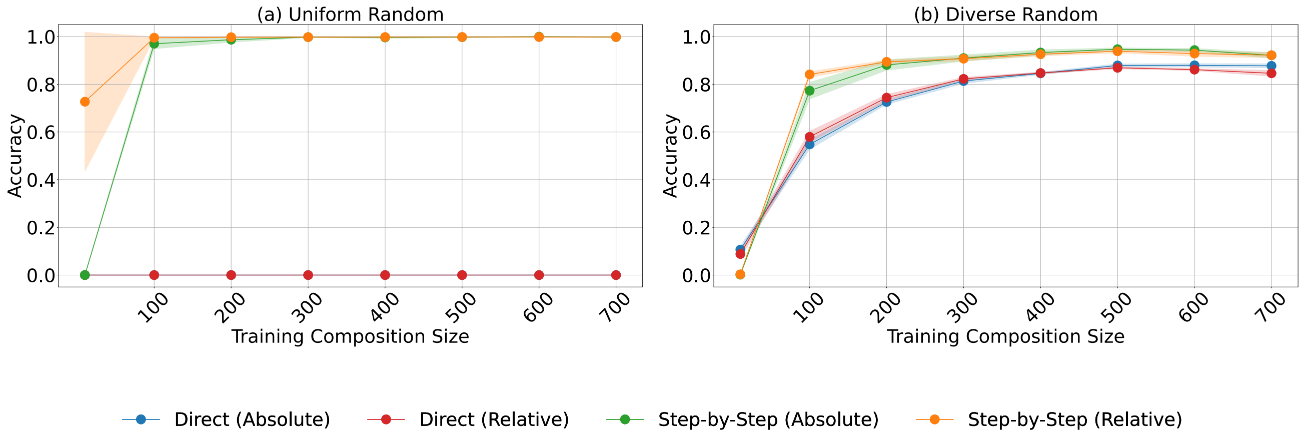

Figure 27. **Effect of number of unique training sequential compositions (randomly sampled, K=6):**: We observe that (a) step-by-step models with relative global embedding need fewer training compositions to generalize faster than absolute embedding models, and (b) For the diverse benchmark, as compared to direct models, step-by-step models need fewer unique training compositions due to extra information available using intermediate outputs.

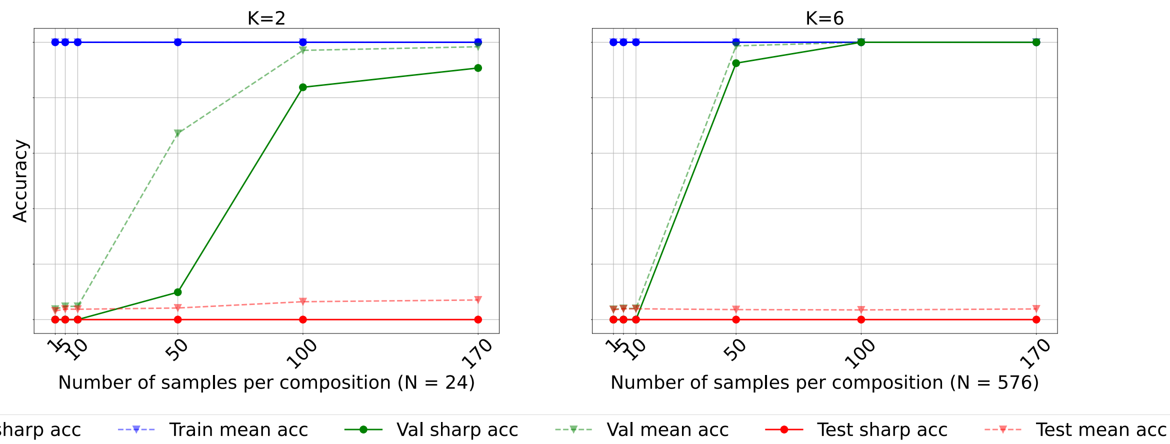

Figure 28. **Effect of number of samples per composition (uniform benchmark, direct model):**. Number of training compositions $N = 24$ for $K = 2$, and $N = 576$ for $K = 6$. We vary the number of input-output samples per training composition in this experiment. We observe that with a small number of training samples, the train/val gap is large, and the model struggles to generalize to training compositions with novel inputs. The train-val gap reduces as the number of samples increases. Test accuracies remain 0%, as expected.

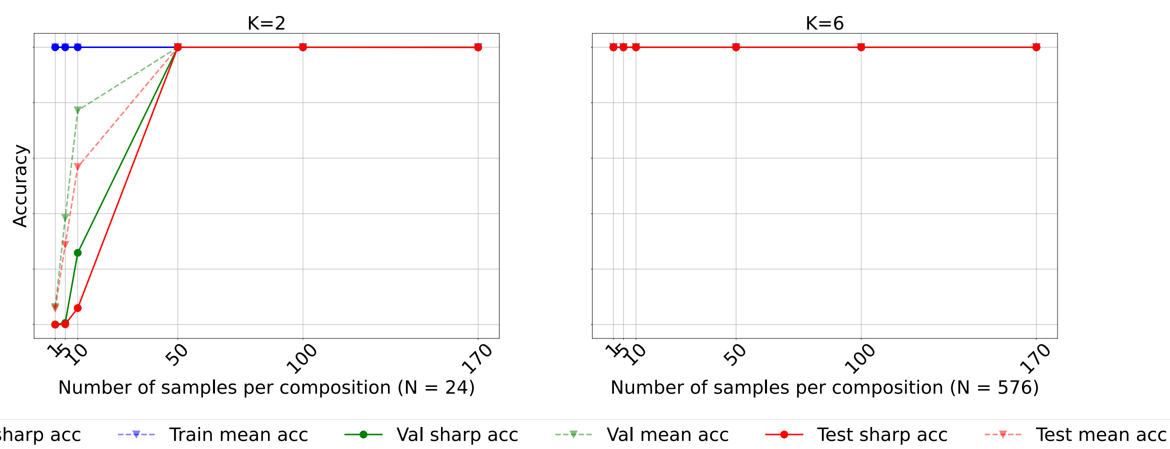

Figure 29. **Effect of number of samples per composition (uniform benchmark, step-by-step model):** We observe that step-by-step models are more sample efficient than direct models due to access to intermediate outputs providing extra information.

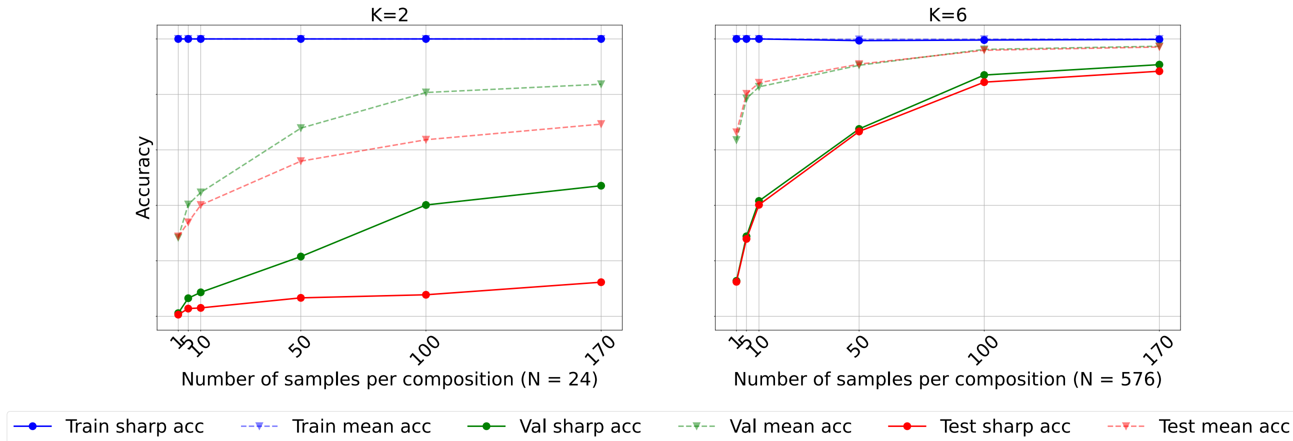

Figure 30. **Effect of number of samples per composition (diverse benchmark, direct model):** For the diverse benchmark, direct models are less sample efficient than the uniform benchmark to generalize on the validation set, especially for K= 2. We observe significant compositional generalization behavior, as seen in Figure 2(b) for large samples, especially for K=6.

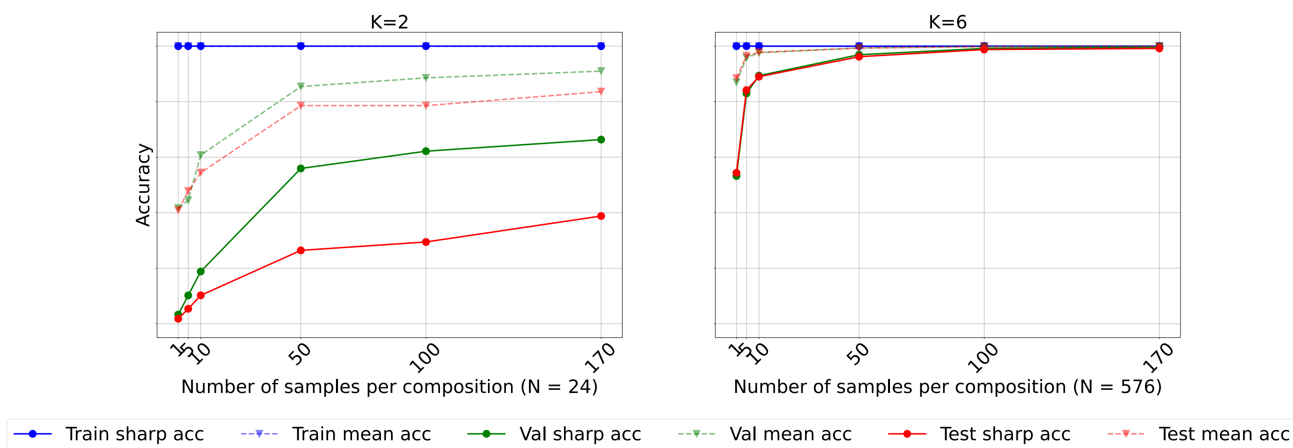

Figure 31. **Effect of number of samples per composition (diverse benchmark, step-by-step model):** Step-by-step models are more sample efficient than direct models, and a similar difference between uniform and diverse benchmarks for K=2 is observed as for direct models.